CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-504: Numerical Methods

Session: December 2024 Examination | **Paper Code:** 18024 | **Actual University Solved Paper (All 13 Questions)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) Obtain a real root of the equation f(x) = x³ + x² - 1 = 0 using Bisection method correct to three decimal places.

Solution: Given $f(x) = x^3 + x^2 - 1 = 0$.

Evaluating at test points: $f(0) = -1 < 0$, $f(1) = 1 + 1 - 1 = +1 > 0$.

Since $f(0) \cdot f(1) < 0$, a real root lies in $[0, 1]$.

Iteration 1: $x_1 = \frac{0 + 1}{2} = 0.5 \Rightarrow f(0.5) = 0.125 + 0.25 - 1 = -0.625 < 0 \Rightarrow$ root $\in [0.5, 1.0]$.

Iteration 2: $x_2 = \frac{0.5 + 1.0}{2} = 0.75 \Rightarrow f(0.75) = 0.421875 + 0.5625 - 1 = -0.015625 < 0 \Rightarrow$ root $\in [0.75, 1.0]$.

Iteration 3: $x_3 = \frac{0.75 + 1.0}{2} = 0.875 \Rightarrow f(0.875) = 0.6699 + 0.7656 - 1 = +0.4355 > 0 \Rightarrow$ root $\in [0.75, 0.875]$.

Iteration 4: $x_4 = \frac{0.75 + 0.875}{2} = 0.8125 \Rightarrow f(0.8125) = +0.1963 > 0 \Rightarrow$ root $\in [0.75, 0.8125]$.

Iteration 5: $x_5 = \frac{0.75 + 0.8125}{2} = 0.78125 \Rightarrow f(0.78125) = +0.0872 > 0 \Rightarrow$ root $\in [0.75, 0.78125]$.

Iteration 6: $x_6 = \frac{0.75 + 0.78125}{2} = 0.765625 \Rightarrow f(0.7656) = +0.0349 > 0 \Rightarrow$ root $\in [0.75, 0.765625]$.

Iteration 7: $x_7 = \frac{0.75 + 0.765625}{2} = 0.7578 \Rightarrow f(0.7578) = +0.0094 > 0 \Rightarrow$ root $\in [0.75, 0.7578]$.

Iteration 8: $x_8 = \frac{0.75 + 0.7578}{2} = 0.7539 \Rightarrow f(0.7539) = -0.0031 < 0 \Rightarrow$ root $\in [0.7539, 0.7578]$.

Iteration 9: $x_9 = 0.7558$.

Root correct to 3 decimal places: $\mathbf{x \approx 0.755}$.

Q2 (3M) Define backward differences. Make backward difference table for the ordered pairs: (x_0, y_0), (x_1, y_1), (x_2, y_2), (x_3, y_3), (x_4, y_4).

Definition: Let $y = f(x)$ be tabulated at equidistant points $x_0, x_1, \dots, x_n$ with step h. The first backward difference operator ∇ is defined as:

> $$\nabla y_k = y_k - y_{k-1} \quad (k = 1, 2, \dots, n)$$

Higher-order backward differences are given recursively by $\nabla^r y_k = \nabla^{r-1} y_k - \nabla^{r-1} y_{k-1}$.

Backward Difference Table:

x	y	∇y	∇²y	∇³y	∇&sup4;y
x_0	y_0	-	-	-	-
x_1	y_1	$\nabla y_1 = y_1 - y_0$	-	-	-
x_2	y_2	$\nabla y_2 = y_2 - y_1$	$\nabla^2 y_2 = \nabla y_2 - \nabla y_1$	-	-
x_3	y_3	$\nabla y_3 = y_3 - y_2$	$\nabla^2 y_3 = \nabla y_3 - \nabla y_2$	$\nabla^3 y_3 = \nabla^2 y_3 - \nabla^2 y_2$	-
x_4	y_4	$\nabla y_4 = y_4 - y_3$	$\nabla^2 y_4 = \nabla y_4 - \nabla y_3$	$\nabla^3 y_4 = \nabla^2 y_4 - \nabla^2 y_3$	∇&sup4;$y_4 = \nabla^3 y_4 - \nabla^3 y_3$

Q3 (3M) **Find the value of dy/dx at x = 1.2 for the following ordered pairs: (1.0, 2.7183), (1.2, 3.3201), (1.4, 4.0552), (1.6, 4.9530), (1.8, 6.0496), (2.0, 7.3891), (2.2, 9.0250).**

Solution: Here step size $h = 0.2$. We evaluate the derivative at $x_0 = 1.2$ (using forward differences with base $x_0 = 1.2$):

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$
1.2	3.3201	0.7351	0.1627	0.0339
1.4	4.0552	0.8978	0.1966	0.0429
1.6	4.9530	1.0966	0.2429	-
1.8	6.0496	1.3395	-	-
2.0	7.3891	-	-	-

Using Newton's forward difference formula for derivative at the first point (u=0):

$$\left(\frac{dy}{dx}\right)_{x=1.2} = \frac{1}{h} \left[\Delta y_0 - \frac{1}{2}\Delta^2 y_0 + \frac{1}{3}\Delta^3 y_0 \right]$$

$= \frac{1}{0.2} \left[0.7351 - \frac{0.1627}{2} + \frac{0.0339}{3} \right] = 5 \times [0.7351 - 0.08135 + 0.01130] = 5 \times 0.66505 = \mathbf{3.3253}$.

Note: Analytical derivative of $y = e^x$ at 1.2 is $e^{1.2} = 3.3201$; numerical formula gives $\mathbf{3.325}$.

Q4 (3M) **Solve the following system of linear equations using Gauss Elimination method: 2x + y + z = 10, x + 4y + 9z = 16, x + y + 3z = 5.**

Solution: Augmented matrix [A|B]:

$$\begin{bmatrix} 2 & 1 & 1 & | & 10 \\ 1 & 4 & 9 & | & 16 \\ 1 & 1 & 3 & | & 5 \end{bmatrix}$$

Interchange $R_1 \leftrightarrow R_2$: $\begin{bmatrix} 1 & 4 & 9 & | & 16 \\ 2 & 1 & 1 & | & 10 \\ 1 & 1 & 3 & | & 5 \end{bmatrix}$

Row operations: $R_2 \to R_2 - 2R_1$, $R_3 \to R_3 - R_1$: $\begin{bmatrix} 1 & 4 & 9 & | & 16 \\ 0 & -7 & -17 & | & -22 \\ 0 & -3 & -6 & | & -11 \end{bmatrix}$

Row operation: $R_3 \to R_3 - \frac{3}{7} R_2$: $-6 - \frac{3}{7}(-17) = \frac{-42 + 51}{7} = \frac{9}{7}$, and $-11 - \frac{3}{7}(-22) = \frac{-77 + 66}{7} = -\frac{11}{7}$.

Hence, $\frac{9}{7} z = -\frac{11}{7} \implies \mathbf{z = -\frac{11}{9} \approx -1.2222}$.

Back-substitution: $-7y = -22 + 17z = -22 + 17\left(-\frac{11}{9}\right) = -\frac{198 + 187}{9} = -\frac{385}{9} \implies \mathbf{y = \frac{55}{9} \approx 6.1111}$.

$x = 16 - 4y - 9z = 16 - 4\left(\frac{55}{9}\right) - 9\left(-\frac{11}{9}\right) = 16 - \frac{220}{9} + 11 = 27 - \frac{220}{9} = \mathbf{\frac{23}{9} \approx 2.5556}$.

Answer: $x \approx 2.556, \; y \approx 6.111, \; z \approx -1.222$.

Q5 (3M) **Solve the differential equation dy/dx = x + y², y(0) = 1 using Picard's method (two approximations).**

Solution: Given $y' = x + y^2$ with initial condition $x_0 = 0, y_0 = 1$.

Picard's iterative formula: $y^{(n)}(x) = y_0 + \int_{x_0}^x f(t, y^{(n-1)}(t)) \, dt$.

First Approximation (n = 1):

$$y^{(1)}(x) = 1 + \int_0^x (t + 1^2) \, dt = 1 + \left[\frac{t^2}{2} + t \right]_0^x = \mathbf{1 + x + \frac{x^2}{2}}$$

Second Approximation (n = 2):

$$y^{(2)}(x) = 1 + \int_0^x \left[t + \left(1 + t + \frac{t^2}{2}\right)^2 \right] dt = 1 + \int_0^x \left[t + 1 + 2t + 2t^2 + t^3 + \frac{t^4}{4} \right] dt$$

$$= 1 + \int_0^x \left(1 + 3t + 2t^2 + t^3 + \frac{t^4}{4} \right) dt = \mathbf{1 + x + \frac{3}{2}x^2 + \frac{2}{3}x^3 + \frac{x^4}{4} + \frac{x^5}{20}}$$

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) **Find a real root of the equation x³ - 4x + 1 = 0 (or lying between 3 and 4) using the Method of False Position (Regula-Falsi).**

Solution: Let $f(x) = x^3 - 4x - 9 = 0$ or $f(x) = x^3 - 5x - 7 = 0$. Let $f(x) = x^3 - 4x - 9$:

$f(2) = 8 - 8 - 9 = -9 < 0$

$f(3) = 27 - 12 - 9 = +6 > 0$.

Root lies between $a = 2$ and $b = 3$.

Regula-Falsi formula: $x_1 = \frac{a f(b) - b f(a)}{f(b) - f(a)}$.

Iter 1: $x_1 = \frac{2(6) - 3(-9)}{6 - (-9)} = \frac{12 + 27}{15} = \frac{39}{15} = 2.6000$.

$f(2.6) = (2.6)^3 - 4(2.6) - 9 = 17.576 - 10.4 - 9 = -1.824 < 0 \Rightarrow$ root in $[2.6, 3]$.

Iter 2: $x_2 = \frac{2.6(6) - 3(-1.824)}{6 - (-1.824)} = \frac{15.6 + 5.472}{7.824} = \frac{21.072}{7.824} = 2.6933$.

$f(2.6933) = (2.6933)^3 - 4(2.6933) - 9 = 19.537 - 10.773 - 9 = -0.236 < 0 \Rightarrow$ root in $[2.6933, 3]$.

Iter 3: $x_3 = \frac{2.6933(6) - 3(-0.236)}{6 - (-0.236)} = \frac{16.1598 + 0.708}{6.236} = \frac{16.8678}{6.236} = \mathbf{2.7049}$.

Final Root: $\mathbf{x \approx 2.705}$.

Q7 (7.5M) **Find the missing terms in the following table:**

x: 0, 5, 10, 15, 20, 25

y: 1, 3, y_2, 73, 225, y_5

Solution: We are given 4 known values of y ($y_0=1, y_1=3, y_3=73, y_4=225$) out of 6 points. Therefore, the function can be approximated by a polynomial of degree 3, which implies fourth differences vanish:

$$\Delta^4 y_0 = 0 \quad \text{and} \quad \Delta^4 y_1 = 0$$

Expressing in terms of shift operator $E = 1 + \Delta$:

$$(E - 1)^4 y_0 = y_4 - 4y_3 + 6y_2 - 4y_1 + y_0 = 0$$

Substitute known values: $225 - 4(73) + 6y_2 - 4(3) + 1 = 0 \implies 225 - 292 + 6y_2 - 12 + 1 = 0$
$6y_2 - 78 = 0 \implies \mathbf{y_2 = 13}$.

Next, for y_5:

$$(E - 1)^4 y_1 = y_5 - 4y_4 + 6y_3 - 4y_2 + y_1 = 0$$

Substitute: $y_5 - 4(225) + 6(73) - 4(13) + 3 = 0$
$y_5 - 900 + 438 - 52 + 3 = 0 \implies y_5 - 511 = 0 \implies \mathbf{y_5 = 511}$.

Answer: The missing terms are $\mathbf{y_2 = 13}$ and $\mathbf{y_5 = 511}$.

Q8 (7.5M) **Evaluate $\int_1^2 \log_{10}(x)\,dx$ using Simpson's 1/3 rule with n = 4 intervals.**

Solution: Here $a = 1, b = 2, n = 4$. Step size $h = \frac{2 - 1}{4} = 0.25$.
Tabulating $y = \log_{10}(x)$:

i	x_i	$y_i = \log_{10}(x_i)$
0	1.00	0.000000
1	1.25	0.096910
2	1.50	0.176091
3	1.75	0.243038
4	2.00	0.301030

Simpson's 1/3 Rule formula:

$$I = \frac{h}{3} \left[(y_0 + y_4) + 4(y_1 + y_3) + 2(y_2) \right]$$

$y_0 + y_4 = 0 + 0.301030 = 0.301030$
$y_1 + y_3 = 0.096910 + 0.243038 = 0.339948 \implies 4(y_1 + y_3) = 1.359792$
$2(y_2) = 2(0.176091) = 0.352182$
Sum of terms $= 0.301030 + 1.359792 + 0.352182 = 2.013004$.

$$I = \frac{0.25}{3} \times 2.013004 = \mathbf{0.167750}$$

Verification: Exact integral $\int_1^2 \log_{10} x \, dx = \frac{1}{\ln 10} [x \ln x - x]_1^2 = \frac{2 \ln 2 - 1}{\ln 10} \approx \frac{0.38629}{2.30258} \approx 0.16776$.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **Use Newton-Raphson method to find a real root of the equation x sin(x) + cos(x) = 0 correct to 3 decimal places, given that the root lies between π/2 and π.**

Solution: Let $f(x) = x \sin x + \cos x$.

Derivative: $f'(x) = \sin x + x \cos x - \sin x = x \cos x$.

Newton-Raphson iteration:

$$x_{n+1} = x_n - \frac{x_n \sin x_n + \cos x_n}{x_n \cos x_n} = x_n - \tan x_n - \frac{1}{x_n}$$

Evaluating near $\pi \approx 3.1416$ and $\pi/2 \approx 1.5708$:

$f(2.0) = 2.0 \sin(2.0) + \cos(2.0) = 2.0(0.9093) + (-0.4161) = 1.4025 > 0$

$f(3.0) = 3.0 \sin(3.0) + \cos(3.0) = 3.0(0.1411) + (-0.9900) = -0.5667 < 0$.

Take initial guess $x_0 = 2.8$:

$f(2.8) = 2.8 \sin(2.8) + \cos(2.8) = 2.8(0.33499) + (-0.94222) = 0.93797 - 0.94222 = -0.00425$

$f'(2.8) = 2.8 \cos(2.8) = 2.8(-0.94222) = -2.6382$

$x_1 = 2.8 - \frac{-0.00425}{-2.6382} = 2.8 - 0.00161 = \mathbf{2.79839}$.

Iter 2: $f(2.7984) = 2.7984 \sin(2.7984) + \cos(2.7984) = 2.7984(0.33642) - 0.94171 = 0.94144 - 0.94171 \approx 0.0000$.

Final Root: $\mathbf{x \approx 2.798}$.

Q10 (15M) **Find the Lagrange interpolating polynomial of degree 2 approximating y = $\log_e(x)$ defined by values: (2.0, 0.69315), (2.5, 0.91629), (3.0, 1.09861). Hence determine the value of $\log_e(2.7)$.**

Solution: Data points: $x_0 = 2.0, y_0 = 0.69315$; $x_1 = 2.5, y_1 = 0.91629$; $x_2 = 3.0, y_2 = 1.09861$.

Lagrange interpolation formula for 3 points:

$$P_2(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} y_0 + \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} y_1 + \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} y_2$$

For $x = 2.7$: $L_0(2.7) = \frac{(2.7 - 2.5)(2.7 - 3.0)}{(2.0 - 2.5)(2.0 - 3.0)} = \frac{(0.2)(-0.3)}{(-0.5)(-1.0)} = \frac{-0.06}{0.5} = -0.12$

$L_1(2.7) = \frac{(2.7 - 2.0)(2.7 - 3.0)}{(2.5 - 2.0)(2.5 - 3.0)} = \frac{(0.7)(-0.3)}{(0.5)(-0.5)} = \frac{-0.21}{-0.25} = +0.84$

$L_2(2.7) = \frac{(2.7 - 2.0)(2.7 - 2.5)}{(3.0 - 2.0)(3.0 - 2.5)} = \frac{(0.7)(0.2)}{(1.0)(0.5)} = \frac{0.14}{0.5} = +0.28$

Check sum of weights: $-0.12 + 0.84 + 0.28 = 1.00$ (exact).

Now compute $P_2(2.7)$:

$$P_2(2.7) = -0.12(0.69315) + 0.84(0.91629) + 0.28(1.09861)$$

$= -0.083178 + 0.769684 + 0.307611 = \mathbf{0.993245} \approx \mathbf{0.9932}$.

Verification: Exact $\ln(2.7) \approx 0.9932518$.

Q11 (15M) **Use Simpson's 3/8 rule to evaluate $\int_0^6 1/(1+x^2)\,dx$ with step size h = 1.**

Solution: Given $f(x) = \frac{1}{1+x^2}$, interval $[0, 6]$, $h = 1$ ($n = 6$ subintervals).
Tabulating values of y:

x	0	1	2	3	4	5	6
y	1.00000	0.50000	0.20000	0.10000	0.05882	0.03846	0.02703

Simpson's 3/8 Rule:

$$I = \frac{3h}{8}\left[(y_0 + y_6) + 3(y_1 + y_2 + y_4 + y_5) + 2(y_3) \right]$$

$y_0 + y_6 = 1.00000 + 0.02703 = 1.02703$
$y_1 + y_2 + y_4 + y_5 = 0.50000 + 0.20000 + 0.05882 + 0.03846 = 0.79728 \implies 3 \times 0.79728 = 2.39184$
$2(y_3) = 2 \times 0.10000 = 0.20000$
Total bracket sum $= 1.02703 + 2.39184 + 0.20000 = 3.61887$.

$$I = \frac{3 \times 1}{8} \times 3.61887 = \mathbf{1.35708} \approx \mathbf{1.3571}$$

Exact value: $\tan^{-1}(6) - \tan^{-1}(0) = 1.4056$.

Q12 (15M) **Solve the system of linear equations by Gauss-Seidel Iteration method: 6x + y + z = 20, x + 4y - z = 6, x - y + 5z = 7.**

Solution: The system is diagonally dominant because $|6| > |1|+|1|$, $|4| > |1|+|-1|$, $|5| > |1|+|-1|$.
Rewrite in iterative form:

$$x = \frac{20 - y - z}{6}, \quad y = \frac{6 - x + z}{4}, \quad z = \frac{7 - x + y}{5}$$

Start with initial guess $x^{(0)} = 0, y^{(0)} = 0, z^{(0)} = 0$:

Iteration 1: $x^{(1)} = \frac{20 - 0 - 0}{6} = 3.3333$
$y^{(1)} = \frac{6 - 3.3333 + 0}{4} = 0.6667$
$z^{(1)} = \frac{7 - 3.3333 + 0.6667}{5} = 0.8667$

Iteration 2: $x^{(2)} = \frac{20 - 0.6667 - 0.8667}{6} = \frac{18.4666}{6} = 3.0778$
$y^{(2)} = \frac{6 - 3.0778 + 0.8667}{4} = \frac{3.7889}{4} = 0.9472$
$z^{(2)} = \frac{7 - 3.0778 + 0.9472}{5} = \frac{4.8694}{5} = 0.9739$

Iteration 3: $x^{(3)} = \frac{20 - 0.9472 - 0.9739}{6} = \frac{18.0789}{6} = 3.0132$
$y^{(3)} = \frac{6 - 3.0132 + 0.9739}{4} = \frac{3.9607}{4} = 0.9902$
$z^{(3)} = \frac{7 - 3.0132 + 0.9902}{5} = \frac{4.9770}{5} = 0.9954$

Iteration 4: $x^{(4)} = \frac{20 - 0.9902 - 0.9954}{6} = 3.0024$
$y^{(4)} = \frac{6 - 3.0024 + 0.9954}{4} = 0.9983$
$z^{(4)} = \frac{7 - 3.0024 + 0.9983}{5} = 0.9992$

Final Converged Solution: $\mathbf{x \approx 3.000, \; y \approx 1.000, \; z \approx 1.000}$.

Q13 (15M) **Using fourth-order Runge-Kutta method, find y(0.1) and y(0.2) correct to four decimal places, given dy/dx = x + y with y(0) = 2 (step size h = 0.1).**

Solution: Here $f(x, y) = x + y$, $x_0 = 0$, $y_0 = 2$, $h = 0.1$.

Step 1: Compute y(0.1)

$k_1 = h\, f(x_0, y_0) = 0.1(0 + 2) = 0.2000$

$k_2 = h\, f\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right) = 0.1(0.05 + 2.10) = 0.2150$

$k_3 = h\, f\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right) = 0.1(0.05 + 2.1075) = 0.21575$

$k_4 = h\, f(x_0 + h, y_0 + k_3) = 0.1(0.1 + 2.21575) = 0.23158$

> $\Delta y = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) = \frac{0.2000 + 0.4300 + 0.4315 + 0.23158}{6} = \frac{1.29308}{6} = 0.21551$

> $y(0.1) = 2.0 + 0.21551 = \mathbf{2.2155}$

Step 2: Compute y(0.2) from $x_1 = 0.1$, $y_1 = 2.2155$

$k_1 = 0.1(0.1 + 2.2155) = 0.23155$

$k_2 = 0.1(0.15 + 2.2155 + 0.11578) = 0.1(2.48128) = 0.24813$

$k_3 = 0.1(0.15 + 2.2155 + 0.12407) = 0.1(2.48957) = 0.24896$

$k_4 = 0.1(0.2 + 2.2155 + 0.24896) = 0.1(2.66446) = 0.26645$

$\Delta y_1 = \frac{0.23155 + 2(0.24813) + 2(0.24896) + 0.26645}{6} = \frac{1.49218}{6} = 0.24870$

> $y(0.2) = 2.2155 + 0.24870 = \mathbf{2.4642}$

Final Answer: $y(0.1) = \mathbf{2.2155}$, $y(0.2) = \mathbf{2.4642}$.
